



3-Bit Counter with PLL and Generalized Clock Divider

An FPGA Clock Generation and Division Exercise

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1. Introduction

This report documents an extension of the FPGA counter labs (Lab 1 and Lab 2), implemented on the ZedBoard (Xilinx Zynq-7000, xc7z020clg484), where a 3-bit counter is driven by a slow clock derived from the board's 100 MHz oscillator. The clock passes through a PLL to produce 8 MHz, then through a clock divider that generalizes to both even and odd division factors, finally reaching 1 Hz so that the counter's transitions can be seen directly on the FPGA LEDs.

For simulation purposes, the divider target is set to 1 MHz instead of 1 Hz, since simulating a full 1 Hz period would take an impractical amount of simulation time. The same divider logic is used for both cases; only the divide value changes.

2. Clock Chain Overview

The overall clock path is:

100 MHz Oscillator → PLL (8 MHz) → Clock Divider → 1 Hz (1 MHz in simulation) → 3-bit Counter → LEDs[2:0]

The PLL is generated using the Xilinx Clocking Wizard IP, the same flow used in Lab 2, configured for a 100 MHz input and an 8 MHz output clock. The clock divider and the 3-bit counter are written as plain synchronous Verilog modules and instantiated in a top-level wrapper, following the same project structure used in Lab 1 and Lab 2.

3. Design Description

3.1 PLL (Clock Wizard)

The Clocking Wizard IP is configured in PLL mode with a 100 MHz primary input clock (clk_in1) and a single output clock (clk_out1) set to 8 MHz. The locked output is used to hold the rest of the design in reset until the PLL output is stable, exactly as done for the counter in Lab 2.

3.2 Generalized Clock Divider

A simple toggle-based divider only works cleanly for even division ratios. To support both even and odd ratios from a single, reusable module, the divider instead uses a free-running counter that compares against a DIVIDE parameter:

The counter increments every cycle of the 8 MHz clock and resets to zero once it reaches (DIVIDE - 1). The divided clock output is toggled once per full count cycle. For an even DIVIDE value this produces a clean 50% duty cycle; for an odd DIVIDE value the output is still a valid clock but with a duty cycle close to, but not exactly, 50%, which is acceptable for this application since only the transition edges matter for counting.

Setting DIVIDE = 8,000,000 divides the 8 MHz PLL output down to 1 Hz for the real FPGA. Setting DIVIDE = 8 divides it down to 1 MHz, which is used purely for simulation so that the counter's behavior can be observed in a reasonable simulation time.

3.3 3-bit Counter

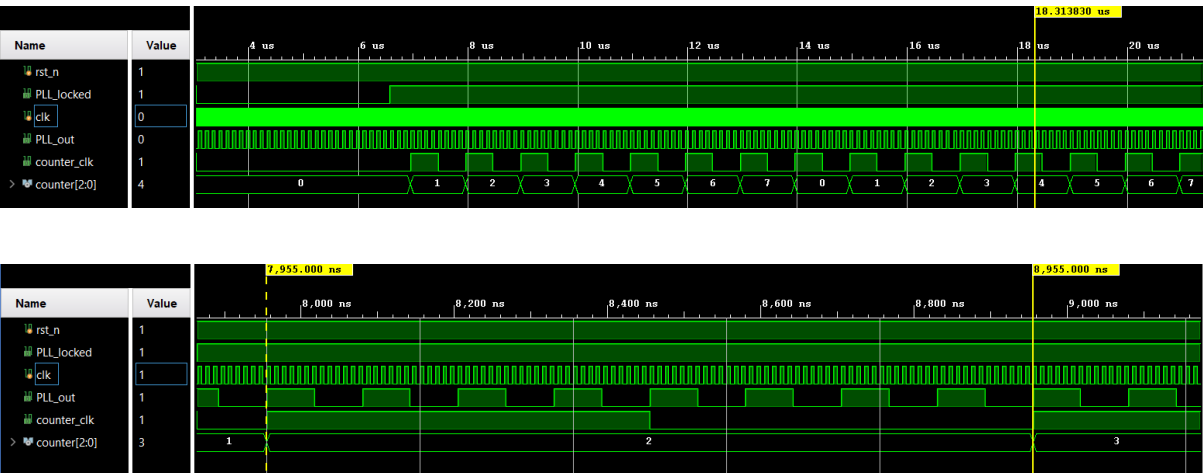
The counter itself is unchanged in structure from Lab 1, except that it is now only 3 bits wide and clocked by the divided clock instead of the raw oscillator or PLL output. It increments by one on every rising edge of the divided clock and wraps around from 3'b111 back to 3'b000. Its output drives

3 of the board's LEDs, so each counter transition is visible to the eye once the clock has been divided down to 1 Hz.

4. Simulation Results

Behavioral simulation was run with the divider set to its simulation value (1 MHz) rather than the real 1 Hz target. The results confirm that:

- The PLL locks and produces a stable 8 MHz clock before the rest of the design leaves reset.
- The clock divider correctly produces one output transition every DIVIDE input cycles, for both an even and an odd DIVIDE value.
- The 3-bit counter increments by exactly one on every edge of the divided clock and wraps correctly from 111 back to 000.

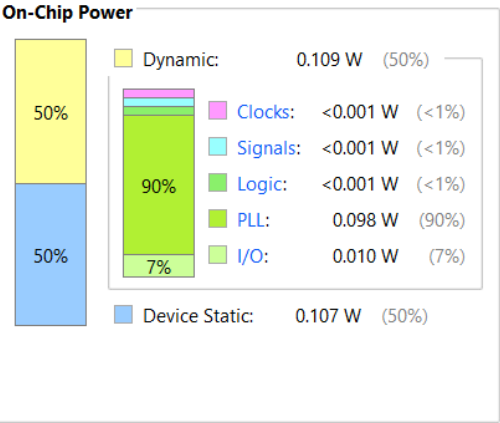


5. Implementation and Reports

Power analysis from Implemented netlist. Activity derived from constraints files, simulation files or vectorless analysis.

Total On-Chip Power: 0.216 W
Design Power Budget: Not Specified
Power Budget Margin: N/A
Junction Temperature: 27.5°C
Thermal Margin: 57.5°C (4.8 W)
Effective θ_{JA} : 11.5°C/W
Power supplied to off-chip devices: 0 W
Confidence level: Low

[Launch Power Constraint Advisor](#) to find and fix invalid switching activity



Design Timing Summary

Setup	Hold	Pulse Width
Worst Negative Slack (WNS): 118.418 ns	Worst Hold Slack (WHS): 0.238 ns	Worst Pulse Width Slack (WPWS): 2.633 ns
Total Negative Slack (TNS): 0.000 ns	Total Hold Slack (THS): 0.000 ns	Total Pulse Width Negative Slack (TPWS): 0.000 ns
Number of Failing Endpoints: 0	Number of Failing Endpoints: 0	Number of Failing Endpoints: 0
Total Number of Endpoints: 3	Total Number of Endpoints: 3	Total Number of Endpoints: 9
All user specified timing constraints are met.		

6. Conclusion

This design builds on Lab 1 and Lab 2 by adding a single, reusable clock divider module that generalizes to any division factor, even or odd, rather than only supporting even values. Combined with the PLL, this produces a clean path from the 100 MHz board oscillator down to a 1 Hz clock suitable for visually observing a 3-bit counter on the FPGA LEDs, while a faster 1 MHz setting of the same module keeps simulation runs short.